Pair Coulomb Density Matrix on a Sphere

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We tabulate the diagonal pair Coulomb density matrix on a sphere.

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CONTENTS

I. INTRODUCTION

We want to find the Pair Coulomb Density Matrix (PCDM) on a Sphere so to extend the work of Ref. [1–3].

We will denote with a Greek index a coordinate basis component and with a Roman index a particle label. We will adopt Einstein summation convention to tacitly assume a sum over repeated Greek indexes. We distinguish between upper contravariant Greek indexes and lower covariant Greek indexes. Passing between the two is accomplished with a contraction with the metric tensor, as usual.

The metric tensor of the Sphere of radius a is

$$||g_{\alpha\beta}|| = a^2 \begin{pmatrix} 1 & 0 \\ 0 & \cos^2\theta \end{pmatrix}. \tag{1.1}$$

with $g = \det||g_{\alpha\beta}|| = a^4\cos^2\theta$.

It is important to realize that on a curved surface the two body problem is generally not separable into a center of mass and a relative one body problem. Therefore, in full generality, we will study the position representation $\rho(\boldsymbol{r}_1,\boldsymbol{r}_2|\boldsymbol{r}_1',\boldsymbol{r}_2';t)$ of the density matrix $\rho=\exp(-Ht)$, where H is the Hamiltonian operator for the two bodies. These are initially at $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ and at an imaginary time t later they are at $\boldsymbol{r}_1'$ and $\boldsymbol{r}_2'$.

Therefore, the PCDM on the Sphere will satisfy the Bloch equation

$$\frac{\partial\rho(\boldsymbol{r}_1,\boldsymbol{r}_2|\boldsymbol{r}_1',\boldsymbol{r}_2';t)}{\partial t} = \left(\frac{\lambda_1}{\sqrt{g_1}}\nabla_{1\alpha}\sqrt{g_1}\nabla_1^\alpha + \frac{\lambda_2}{\sqrt{g_2}}\nabla_{2\alpha}\sqrt{g_2}\nabla_2^\alpha - \phi(\boldsymbol{r}_1,\boldsymbol{r}_2)\right)\rho(\boldsymbol{r}_1,\boldsymbol{r}_2|\boldsymbol{r}_1',\boldsymbol{r}_2';t), \tag{1.2}$$

$$\rho(\boldsymbol{r}_1,\boldsymbol{r}_2|\boldsymbol{r}_1',\boldsymbol{r}_2';0) = \delta(\boldsymbol{r}_1-\boldsymbol{r}_1')\delta(\boldsymbol{r}_2-\boldsymbol{r}_2')/\sqrt{g_1 g_2}, \tag{1.3}$$

where, for particle $i=1,2$, $\lambda_i=\hbar^2/2m_i$ with m_i the mass of particle i, $g_i = a^4\sin^2\theta_i$, $\boldsymbol{\nabla}_i = (\nabla_{i1},\nabla_{i2}) = (\partial/\partial\theta_i,\partial/\partial\varphi_i)$ is the gradient respect to the coordinates $\boldsymbol{r}_i=(r_i^1,r_i^2)=(\theta_i,\varphi_i)$, δ is a two dimensional Dirac delta function, ρ is the pair density matrix, and ϕ is the Coulomb pair potential.

The Coulomb pair potential for particles moving **on** the surface can be chosen as

$$\phi_{\rm on}(\boldsymbol{r}_1,\boldsymbol{r}_2) = Z_1Z_2e^2/d(\boldsymbol{r}_1,\boldsymbol{r}_2), \tag{1.4}$$

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where $Z_i e$ is the charge of particle i and for the distance between the two points $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$, $d(\boldsymbol{r}_1, \boldsymbol{r}_2)$, on the Sphere we can either choose the geodesic distance

$$d_g(\boldsymbol{r}_1, \boldsymbol{r}_2) = a \arccos[\sin\theta_1 \sin\theta_2 + \cos\theta_1 \cos\theta_2 \cos(\varphi_1 - \varphi_2)], \tag{1.5}$$

or, viewing the Sphere as embedded in the three dimensional space, the Euclidean distance

$$d_e(\boldsymbol{r}_1, \boldsymbol{r}_2) = 2a \sin(d_g/2a). \tag{1.6}$$

For particles living **in** the surface [4] we may choose

$$\phi_{\rm in}(\boldsymbol{r}_1, \boldsymbol{r}_2) = -Z_1 Z_2 e^2 \ln[d_e(\boldsymbol{r}_1, \boldsymbol{r}_2)/\ell], \tag{1.7}$$

with ℓ a constant with the dimensions of a distance. We will call $\phi^g_{\rm on}$, $\phi^e_{\rm on}$, $\phi_{\rm in}$ the three cases just illustrated.

We will measure energy in units of the Hartree e^2/a_B, with e the charge of the electron and $a_B = \hbar^2/m_e e^2$ Bohr radius, with m_e the mass of the electron. And length in units of the Bohr radius a_B.

The free particles, $\phi \to 0$, pair density matrix can be determined from the work of Bastianelli and Corradini [5]. We will call their solution for the single particle i, $\rho_{i0}(\boldsymbol{r}_i, \boldsymbol{r}'_i; t)$. Therefore the pair free density matrix will be

$$\rho_0(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; t) = \rho_{10}(\boldsymbol{r}_1, \boldsymbol{r}'_1; t)\rho_{20}(\boldsymbol{r}_2, \boldsymbol{r}'_2; t). \tag{1.8}$$

Then our PCDM can be written as

$$\rho(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; t) = \rho_0(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; t) e^{-P(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; t)}, \tag{1.9}$$

where P is the *inter-action*.

Aim of this work is to tabulate $P(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; t)$ for the three Coulomb cases illustrated above, $P^g_{\rm on}$, $P^e_{\rm on}$, $P_{\rm in}$. For small $t \ll 1$, for each of these cases, the *primitive approximation* [6]

$$P_{\rm primitive} = t\phi, \tag{1.10}$$

holds.

Therefore, introducing a timestep τ, we can find the PDCM at all subsequent timesteps through the following convolution

$$\rho(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; n\tau) = \int \rho(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}_{1,1}, \boldsymbol{r}_{2,1}; \tau)\rho(\boldsymbol{r}_{1,1}, \boldsymbol{r}_{2,1}|\boldsymbol{r}_{1,2}, \boldsymbol{r}_{2,2}; \tau)\ldots\rho(\boldsymbol{r}_{1,n-1}, \boldsymbol{r}_{2,n-1}|\boldsymbol{r}'_1, \boldsymbol{r}'_2; \tau) \times \prod_{k=1}^{n-1} \sqrt{g_{1,k} g_{2,k}} d\boldsymbol{r}_{1,k}\boldsymbol{r}_{2,k}, \tag{1.11}$$

where we denote with $\boldsymbol{r}_{i,k}$ the position of particle i at timeslice k, $g_{i,k}$ is the determinant g calculated at $\boldsymbol{r}_{i,k}$, $d\boldsymbol{r} = d\theta d\varphi$, and we choose τ small enough so that

$$\rho(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; \tau) \approx \rho_0(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; \tau) e^{-\tau[\phi(\boldsymbol{r}_1, \boldsymbol{r}_2) + \phi(\boldsymbol{r}'_1, \boldsymbol{r}'_2)]/2} \tag{1.12}$$

$$= \rho_{10}(\boldsymbol{r}_1, \boldsymbol{r}'_1; \tau)\rho_{20}(\boldsymbol{r}_2, \boldsymbol{r}'_2; \tau) e^{-\tau[\phi(\boldsymbol{r}_1, \boldsymbol{r}_2) + \phi(\boldsymbol{r}'_1, \boldsymbol{r}'_2)]/2}, \tag{1.13}$$

is a good approximation. According to the analysis of Ref. [5] we can further approximate

$$g(\boldsymbol{r}_i)^{1/4} \rho_{i0}(\boldsymbol{r}_i, \boldsymbol{r}'_i; \tau) g(\boldsymbol{r}'_i)^{1/4} \approx \exp\left\{-\frac{1}{2\mu_i\tau}\delta_{\alpha\beta}(\boldsymbol{r}_i - \boldsymbol{r}'_i)^\alpha(\boldsymbol{r}_i - \boldsymbol{r}'_i)^\beta - \tau\frac{V_R(x_i) + V_R(x'_i)}{2}\right\}, \tag{1.14}$$

where $\delta_{\alpha\beta}$ is the Kronecker delta, $x_i = \delta_{\alpha\beta}{\boldsymbol{r}_i}^\alpha {\boldsymbol{r}_i}^\beta$, and V_R is the effective potential due to the curvature of the Sphere, $R = 2M^2$ with $M = 1/a$, namely

$$V_R(x) = -\frac{M^2}{6} - \frac{5(Mx)^2 - 3 + [(Mx)^2 + 3]\cos(2Mx)}{48x^2\sin^2(Mx)}, \tag{1.15}$$

which is equation (2.32) of Ref. [5]. We will perform the $4(n-1)$ integrations in Eq. (1.11) through a simple quadrature method.

Even if in general we will be interested in the full $P(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}'_1, \boldsymbol{r}'_2; t)$ in practice in a many body simulation we may use the *end point approximation* for the inter-action [6] where one only needs the diagonal components $P(\boldsymbol{r}_1, \boldsymbol{r}_2|\boldsymbol{r}_1, \boldsymbol{r}_2; t) = P(d(\boldsymbol{r}_1, \boldsymbol{r}_2); t)$. Moreover on a sphere the distance between any two points $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ is bounded from above: the geodesic distance must be smaller than πa and the Euclidean distance than $2a$.

On the diagonal we will have

$$\rho(\boldsymbol{r}_1,\boldsymbol{r}_2|\boldsymbol{r}_1,\boldsymbol{r}_2;t)=\rho(d(\boldsymbol{r}_1,\boldsymbol{r}_2);t)=\rho_{10}(\boldsymbol{0},\boldsymbol{0};t)\rho_{20}(\boldsymbol{0},\boldsymbol{0};t)e^{-P(d(\boldsymbol{r}_1,\boldsymbol{r}_2);t)}, \tag{1.16}$$

and $\ln[2\pi t\sqrt{g(\boldsymbol{0})}\rho_{i0}(\boldsymbol{0},\boldsymbol{0};t)]$ is given up to orders t^8 in Ref. [5] (see their Eq. (3.23)). For completeness we report their perturbative expansion in Eq. (A1) of Appendix A .

We will then tabulate

$$P(m\delta;n\tau)=-\ln\left[\frac{\rho(m\delta;n\tau)}{\rho_0(m\delta;n\tau)}\right], \quad n=1,2,3,\ldots, \quad m=1,2,3\ldots,m_{\rm max} \tag{1.17}$$

for given small τ and δ. For an imaginary time $t_n=n\tau$ this will require a number of $4(n-1)m_{\rm max}$ quadratures to calculate $\rho(m\delta;n\tau)=\rho(d(\boldsymbol{r}_1,\boldsymbol{r}_2);t_n)=\rho(\boldsymbol{r}_1,\boldsymbol{r}_2|\boldsymbol{r}_1,\boldsymbol{r}_2;t_n)$ for $m=1,2,3,\ldots,m_{\rm max}$, through the convolution (1.11). Each convolution requires 2 quadratures over the azimuthal angle $\varphi\in[-\pi,\pi[$ and the polar angle $\theta\in[-\pi/2,\pi/2]$. Choosing an angular discretization $\delta_\varphi=2\delta_\theta$ we will have that each angular integral reduces to a sum over $l=\pi/\delta_\theta$ terms. Then in order to tabulate $P(d;t_n)$ we need $4(n-1)m_{\rm max}l$ functions evaluations. This becomes more computationally demanding at later imaginary times t but in a many body simulation the pair-product inter-action is needed at a fixed small t, which determines the discretization error [6]. So one needs to tabulate it just once, at the beginning of the simulation.

In Table I we show $P(m\delta;n\tau)$ for $a=1,m_1=m_2=m_e,Z_1=-Z_2=-1$, $\tau=0.001,\delta=0.1$ and $n=10$.

TABLE I. Table for $P(m\delta;n\tau)$ for $a=1,m_1=m_2=m_e,Z_1=-Z_2=-1$, $\tau=0.001,\delta=0.1$ and $n=10$. For the quadratures we used Simpson rule with $\delta_\theta=0.01$ so that $l=314$.

$P^g_{\rm on}$	$P^e_{\rm on}$	$P_{\rm in}$	m
			1
			2
			20
			31

II. THE CODE

We give here the FORTRAN code listing to generate the table of Eq. (1.17):

Appendix A: $\rho_{i0}(\boldsymbol{0},\boldsymbol{0};t)$

Calling $\mu_i=m_i/m_e$ and $R=2/a^2$ the Sphere scalar curvature, we find from equation (3.23) of Ref. [5]

$$\begin{aligned}\ln\left[2\pi t\sqrt{g(\boldsymbol{0})}\rho_{i,0}(\boldsymbol{0},\boldsymbol{0};t)\right]=&\frac{tR_i}{12}+\\&\frac{(tR_i)^2}{6!}\frac{1}{2}+\\&\frac{(tR_i)^3}{9!}16+\\&\frac{(tR_i)^4}{10!}\frac{59}{4}+\\&\frac{(tR_i)^5}{11!}\frac{58}{3}+\\&\frac{(tR_i)^6}{13!}\frac{550789}{1260}+\\&\frac{(tR_i)^7}{14!}\frac{46838}{45}+\\&\frac{(tR_i)^8}{17!}\frac{596004707}{720}+\mathscr{O}(t^9),\end{aligned} \tag{A1}$$

where $R_i = R/\mu_i$.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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[1] E. L. Pollock, Properties and computation of the Coulomb pair density matrix, Comput. Phys. Commun. **52**, 49 (1988).
[2] P. Vieillefosse, Coulomb Pair Density Matrix I, J. Stat. Phys. **74**, 1195 (1994).
[3] P. Vieillefosse, Coulomb Pair Density Matrix II, J. Stat. Phys. **80**, 461 (1995).
[4] R. Fantoni, Plasma living in a curved surface at some special temperature, Physica A **177**, 524 (2019).
[5] F. Bastianelli and O. Corradini, On the simplified path integral on spheres, Eur. Phys. J. C **77**, 731 (2017).
[6] D. M. Ceperley, Path integrals in the theory of condensed Helium, Rev. Mod. Phys. **67**, 279 (1995).